Benchmarking Dynamic Frequency Estimators for Low-Inertia IBR Grids: A Latency–Robustness Trade-off Analysis

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***Abstract*—In low-inertia grids dominated by Inverter-Based Resources (IBRs), the latency and robustness of frequency estimation are critical for protection and dynamic security assessment. This paper presents a stress-oriented benchmark of twelve estimators from five algorithmic families evaluated under five scenarios using dual-rate numerical simulation (1 MHz physics / 10 kHz DSP). Each estimator is independently tuned per scenario via grid search—establishing the performance ceiling for each estimator–scenario pair—and assessed on RMSE, peak error, trip-risk duration (T_{trip}), and computational cost ($N = 30$ Monte Carlo). Tests (A)–(C) follow the IEC/IEEE 60255-118-1 test philosophy; Scenarios (D)–(E) introduce IBR-specific composite disturbances beyond the standard scope. Results show that several estimators passing standard compliance tests degrade substantially under composite IBR disturbances, and that no single family dominates across all scenario types. Window-based and loop-based methods exhibit complementary strengths; data-driven approaches exhibit higher computational cost without consistent accuracy gains over scenario-optimized classical methods in the evaluated scenarios. The findings suggest that the IEC/IEEE 60255-118-1 test scope may not fully characterize estimator resilience in IBR-dominated conditions, and that composite phase-step and harmonic distortion scenarios expose failure modes not revealed by isolated single-disturbance tests.**

***Index Terms*—Frequency Estimation, Inverter-Based Resources (IBR), Low-Inertia Systems, Extended Kalman Filter (EKF), Phase-Locked Loop (PLL), RoCoF Measurement, Sub-Cycle Transient Tracking, Taylor-Fourier Methods, Benchmarking, Protection and False-Trip Risk.**

I. Introduction

The transition toward inverter-dominated power systems is reducing grid inertia and accelerating frequency dynamics. Frequency deviations and Rate of Change of Frequency (RoCoF) can exceed protection thresholds within milliseconds [1], [2]. Unlike synchronous machines, Inverter-Based Resources (IBRs) regulate power electronically, decoupling the primary energy source from electrical frequency and increasing the impact of harmonics, phase discontinuities, and nonstationary disturbances on protection and dynamic security assessment [1], [3], [4]. In IBR-dominated grids, phase discontinuities can induce cycle-slips in phase-locked loops, sustained RoCoF can exceed the tracking capacity of multi-cycle window methods, and converter-driven harmonic injection can cause spectral leakage in DFT-based estimators. These failure modes may occur simultaneously, yet existing benchmarks typically assess them in isolation [5], leaving a gap for protection-class applications.

The main estimator families address the latency–robustness trade-off differently. SRF-PLL is widely used but susceptible to cycle-slip under abrupt phase discontinuities [6], [7]. Window-based methods (IpDFT, TFT) provide strong harmonic rejection at the cost of structural observation latency [8], [9]. Model-based filters (EKF, UKF) improve tracking under transients [10], [11], while data-driven approaches may offer accuracy gains at higher computational cost [5], [12].

This paper presents a stress-oriented benchmark of twelve estimators from five algorithmic families, evaluated in a local, non-time-synchronized relay-class configuration under five scenarios: (A)–(C) follow the IEC/IEEE 60255-118-1 test philosophy; (D)–(E) introduce IBR-specific composite disturbances beyond the standard scope. The main contributions are: (1) a comprehensive benchmark of twelve frequency estimators spanning five algorithmic families under five stress scenarios—including two composite IBR-specific disturbance scenarios (Scenarios D–E) not covered by IEC/IEEE 60255-118-1; (2) scenario-wise grid-search tuning with dual-rate simulation (1 MHz physics / 10 kHz DSP), establishing the performance ceiling for each estimator–scenario pair; and (3) a comparative analysis of RMSE, peak error, trip-risk duration, and computational cost, providing actionable selection guidance for relay-class deployment in low-inertia grids.

A. Algorithmic Landscape and Prior Art

Frequency estimation involves a latency–robustness trade-off [5]. *Structural observation latency* (N_c/f_0 for window-based methods) is signal-independent; *stochastic estimation lag* depends on disturbance type and estimator parameters—these are independent metrics. Window-based methods (IpDFT, TFT) provide harmonic rejection but impose structural latency [8], [9]. Loop-based methods (SRF-PLL, SOGI-FLL) are computationally efficient but susceptible to cycle-slip under phase discontinuities [13]. Adaptive recursive estimators (RLS, VFF-RLS) may diverge under severe transients.

Model-based filters (EKF, UKF, RA-EKF) offer improved tracking [10], [11]. Data-driven approaches (Koopman, PI-GRU) show competitive accuracy at higher cost [5], [12]. Prior studies evaluate disturbances in isolation; composite scenarios are not assessed simultaneously. Table I compares all twelve evaluated estimators.

II. Methodology and Test Framework

A dual-rate simulation framework separates continuous-time system physics ($f_{\mathrm{s,phys}} = 1\,\mathrm{MHz}$) from discrete-time estimation ($f_{\mathrm{s,DSP}} = 10\,\mathrm{kHz}$, ratio 100:1). Throughout the paper, ω denotes angular frequency in rad/s; scalar frequency values are reported in Hz or mHz ($f = \omega/2\pi$). This yields negligible numerical error in ground-truth frequency trajectories while reproducing the sampling constraints of commercial protection relays.

A. Evaluated Estimators

Twelve estimators from five algorithmic families are evaluated (Table I). **Loop-based:** SRF-PLL and a bandpass-pre-filtered SOGI-FLL (not a standard textbook SOGI-FLL [13]). The implemented SOGI-FLL adds two non-standard front-end stages: (i) an IIR bandpass pre-filter centred at 50 Hz to suppress harmonic contamination before the SOGI core (without it, the 5th and 7th harmonics in Scenarios D–E induce false frequency deviations in the FLL loop); and (ii) a FastRMS adaptive amplitude normalizer that compensates voltage-step amplitude variations and maintains the SOGI resonance condition. These additions create a narrow effective passband that suppresses apparent frequency deviation under a phase jump but cannot track sustained linear RoCoF, explaining the divergent ranking across Scenarios B and D. A unit inconsistency in the standard SOGI-FLL adaptation law (rad/s vs. Hz, factor $2\pi \cdot 60 = 120\pi$ rad/s) was identified and corrected prior to evaluation; without the correction the adaptation gain would be mis-scaled by $\approx 377\times$, making the FLL loop either unstable or unresponsive depending on gain direction. The results are therefore not directly comparable to textbook SOGI-FLL performance figures from [13]. **Window-based:** IpDFT with a Hann window and three-point magnitude-parabolic interpolation [15] (the complex Jacobsen estimator is unsuitable here because Hann-window sidelobes are negative-real, which inverts the interpolation direction); TFT with a 2nd-order Taylor-series expansion of the fundamental component, Hann-weighted, which improves RoCoF tracking relative to a static DFT at the cost of higher noise sensitivity [9]. **Adaptive recursive:** RLS and VFF-RLS with a 2nd-order AR(2) bandpass pre-whitening filter. **Data-driven:** Koopman (RK-DPMU) and PI-GRU. **Model-based:** EKF, UKF, and RA-EKF. The column "RoCoF Out." indicates whether RoCoF is available as an explicit output (S) or derived by implicit differentiation (D). The UKF includes a 10-sample output moving average not present in the classic EKF; at $f_{\mathrm{s,DSP}} = 10\,\mathrm{kHz}$ this adds $\approx 0.5\,\mathrm{ms}$ of causal group delay (half the window length) and increases effective settling time by the same amount—a hidden smoothing trade-off that slightly favours the UKF on RMSE but penalises its transient response relative to a raw-output EKF.

1) Implemented RA-EKF Estimator

RoCoF state augmentation in the EKF has been studied previously [10], [11]. The RA-EKF uses a 4-state vector $\mathbf{x}_k = [\theta_k, \omega_k, A_k, \dot{\omega}_k]^\top$ ($\omega = 2\pi f$) with the discrete-time state-transition matrix

$$\mathbf{F} = \begin{bmatrix} 1 & \Delta t & 0 & \frac{1}{2}\Delta t^2 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \tag{1}$$

and scalar measurement $y_k = A_k \cos\theta_k$ with linearised observation row vector

$$\mathbf{H}_k = \big[-A_k \sin\theta_k,\ 0,\ \cos\theta_k,\ 0\big]. \tag{2}$$

Two IBR-tailored mechanisms are added. *Innovation-Driven Covariance Scaling:* $\mathbf{Q}_k = \mathrm{clip}(r_k, 0.25, 4)\,\mathbf{Q}_{\mathrm{base}}$, where $r_k = |\nu_k|/\hat{\sigma}_\nu$ and $\hat{\sigma}_\nu$ is the running estimate of the steady-state innovation standard deviation, updated each step as an exponentially weighted moving average $\hat{\sigma}_\nu \leftarrow 0.99\,\hat{\sigma}_\nu + 0.01\,|\nu_k|$. This inflates process noise during transients and shrinks it at steady state. *Event Gating:* when $|\nu_k| > \gamma\hat{\sigma}_\nu$ ($\gamma = 2.0$, fixed after preliminary sensitivity experiments), a soft re-initialization updates θ_k only, preserving ω_k and $\dot{\omega}_k$ to limit cycle-slip behavior at phase discontinuities. This heuristic trades strict stochastic optimality for bounded nonlinear stability during abrupt phase steps; γ was not optimised and its sensitivity is not further explored here.

B. Test Scenarios and Noise Model

Five scenarios stress the estimators under representative IBR-grid conditions (Fig. 1). The ground-truth frequency is defined as the time derivative of the phase signal, computed analytically from the continuous-time generator model (central difference for numerical evaluation); this definition produces a non-flat frequency trajectory after phase jumps in Scenarios D and E. Tests (A)–(C) use Additive White Gaussian Noise (AWGN) with $\sigma = 0.001$ pu; Scenario (D) uses $\sigma = 0.005$ pu; and Scenario (E) uses $\sigma = 0.003$ pu base noise plus sparse impulsive samples (amplitude ≤ 0.05 pu, probability $< 1\%$).

- **(A) Magnitude Step:** $+10\%$ voltage step with zero phase angle at the disturbance onset.
- **(B) Frequency Ramp:** $+3.0$ Hz/s RoCoF (IEEE 1547-2018 Category III compliant).
- **(C) AM Modulation:** 10% depth at 2 Hz.
- **(D) Composite Islanding:** instantaneous $+60^\circ$ phase jump (maximum mismatch during breaker reclosing [7]), 4% 5th harmonic (250 Hz), 2% 7th harmonic (350 Hz), 0.5% 32.5 Hz interharmonic (a representative sub-synchronous interharmonic in the range produced by IBR converter control interactions [3], [4]; THD $\approx$ 4.5%, IEEE 519 compliant), and $\sigma = 0.005$ pu noise.
- **(E) IBR Multi-Event:** 5 s sequence with harmonic-rich steady state (5th: 4%, 7th: 2%; THD $\approx$ 4.5%, IEEE 519 compliant), a $+40^\circ$ phase jump, a -3.0 Hz/s RoCoF segment reaching a nadir of 57.5 Hz (IEEE 1547-2018 UFLS limit), a $+80^\circ$

TABLE I: Unified Qualitative Benchmark of Frequency Estimators (relay-class, non-synchronized context). Scores (0=Fail, 1=Poor, 2=Fair, 3=Excellent) are qualitative assessments derived from observed benchmark performance (this work; see Tables II–III for quantitative evidence) and published literature [9], [10], [13]; they are intended for algorithmic comparison and are not certification ratings. h_{HU}: Harmonic/Unbalance Immunity; $r_{32.5}$: rejection of 32.5 Hz sub-synchronous interharmonic (representative of IBR converter control interactions [3]; used in Scenarios D–E, Section II-B); n_{sens}: noise floor sensitivity; n_{NG}: Non-Gaussian noise robustness; Latency class: P1 $<$ 20 ms, P2: 20–40 ms, M1 $>$ 40 ms, P3: sub-cycle ($<$1 ms); v_{RT}: real-time viability (3=embedded, 0=GPU only); RoCoF Out.: S = explicit state output, D = implicit differentiation; *Phase Resp.*: phase-jump response type — S = settles without cycle-slip, D = cycle-slip susceptible; †IpDFT latency is structural: $N_c/f_0 = 33$ ms at $N_c = 2$ to 167 ms at $N_c = 10$ (observation delay, not stochastic lag).

Method	Phase Resp.	RoCoF Out.	h_{HU}	$r_{32.5}$	n_{sens}	n_{NG}	Latency Class	v_{RT}	Comp. Order	Dominant Failure Mode
I. SOTA Methods (Post-2020)										
VFF-RLS	S	D	2	3	3	2	P2	3	$O(p^2)$	VFF blow-up
UKF	S	S	2	2	2	2	P1	1	$O(n^3)$	Sigma-pt collapse
Koopman (DPMU)	S	S	2	2	2	1	M1	1	$O(n^2)$	Model mismatch
PI-GRU (DL)	S	S	1	1	2	1	P1	0	$O(\mathrm{Conv})$	Dataset shift
II. Industrial / Advanced Digital										
RA-EKF	S	S	2	2	3	2	P1	1	$O(n^3)$	Jacobian divergence
EKF	S	S	2	2	2	1	P1	3	$O(n^3)$	Phase-jump divergence
TFT	S	S	2	2	2	1	P1	1	$O(N^2)$	Runge overshoot
SRF-PLL	D	D	1	1	1	0	P2	3	$O(1)$	Cycle slip / Lag
SOGI-FLL	D	D	2	1	2	1	P2	3	$O(1)$	Tuning trade-off
IpDFT†	S	D	3	1	3	0	M1	2	$O(N \log N)$	Leakage / Latency
III. Legacy Baselines										
RLS (fixed λ)	S	D	1	1	1	0	P2	3	$O(p^2)$	Covariance blow-up
TKEO	S	D	0	0	0	0	P3	3	$O(1)$	Noise amplification

jump with second-order ring-down recovery, and impulsive noise [11].

(F) Primary Frequency Response (*planned extension*): a second-order electromechanical recovery following a generation loss event, modelling governor–turbine dynamics with damping. Monte Carlo evaluation under this scenario is reserved for future work.

Scenarios (A)–(C) follow the IEC/IEEE 60255-118-1 test philosophy, whereas Scenarios (D)–(E) introduce IBR-specific disturbances beyond the standard scope.

C. Performance Metrics and Optimization Protocol

A scenario-wise grid search (Table II) selects the configuration minimizing RMSE per estimator–scenario pair, establishing the *performance ceiling* of each family under each disturbance type. *Scenario-wise tuning identifies this ceiling; practical deployment requires a fixed parameter set across all disturbance types.* PI-GRU is a fixed pre-trained generalist baseline and is not tuned per scenario; all classical methods are scenario-optimized. Monte Carlo uses $N = 30$ independent runs with fixed tuned parameters (seeds 2000–2029; tuning seed: 42). Let $e(t) = \hat{f}(t) - f_{\mathrm{true}}(t)$. Four metrics are reported: **RMSE**, **Peak Error** ($|e|_{\max}$), **Settling Time** (first time $|e|$ falls and remains ≤ 200 mHz for ≥ 20 ms), and **Trip-Risk Duration** (T_{trip}: cumulative time with $|e| > 0.5$ Hz). *Ground-truth frequency* is the analytical time derivative of the phase signal, computed from the continuous-time generator model. Computational cost is mean time per sample over 20 repetitions (single-threaded Python 3.13, Intel Core Ultra 5 125H). *Python timings are relative benchmarks only; C/FPGA implementations yield 10–100× lower absolute latencies.* IpDFT structural latency: $N_c/f_0 = 33$–167 ms (observation delay, not stochastic lag).

For the non-normative compliance heatmap (Fig. 2g), a *Pass* requires RMSE < 0.05 Hz, $|e|_{\max} < 0.5$ Hz, and $T_{\mathrm{trip}} < 0.1$ s. These thresholds are drawn from IEC/IEEE 60255-118-1 accuracy classes and are used here for visualization only; no formal certification claim is implied. A symbol **F*** denotes *marginal failure*: RMSE exceeds the 0.05 Hz threshold by $< 1.5\%$ or the peak breach is sub-cycle in duration—compliant in all other metrics and considered borderline rather than a hard fail.

III. Simulation Results and Benchmarking

All frequency errors are reported in mHz unless otherwise stated.

A. Performance on Standard IEC/IEEE Scenarios

Table III summarizes performance on standard scenarios (3 s.f.). **Steady-state:** SRF-PLL achieves RMSE = 0.130 mHz (Modulation); window-based methods reach

TABLE II: Tuning Parameter Search Space (≈5,000 total runs). *log*: 12 log-spaced pts/decade; *linear*: 10 pts; $\{\cdot\}$: discrete set. Units: K_p [pu/rad], K_i [pu/(rad·s)], R [pu^2], N_c [cycles], N_{smooth} [samples].

Method	**Param.**	**Search Range (spacing; # pts)**
SRF-PLL	K_p, K_i	$[1, 60]$, $[1, 200]$ (linear, 10 pts each)
EKF / RA-EKF	Q	10^{-4}–10^{4} (log, 12 pts)
	R	10^{-5}–10^{0} (log, 12 pts; $R_{\max} = 1.0$)
RA-EKF only	$\hat{\sigma}_\nu$, γ	$\{0.1, 0.3\}$, 2.0 (fixed; 2 pts / 1 pt)
SOGI-FLL	k, g_{FLL}	$\{0.5, 1.0, 1.5, 2.0\}$, $\{5, 10, \ldots, 150\}$
IpDFT	N_c	$\{2, 3, 4, 6, 8, 10\}$ cycles (6 pts)
RLS / VFF	λ, N_{smooth}	0.9–0.99999 (log, 12 pts), 2–2000 samp. (log, 10 pts)
TFT	N_c	$\{2, 3, 4, 6\}$ cycles (4 pts)
TKEO	Smooth window	10–50 samples (linear, 9 pts)

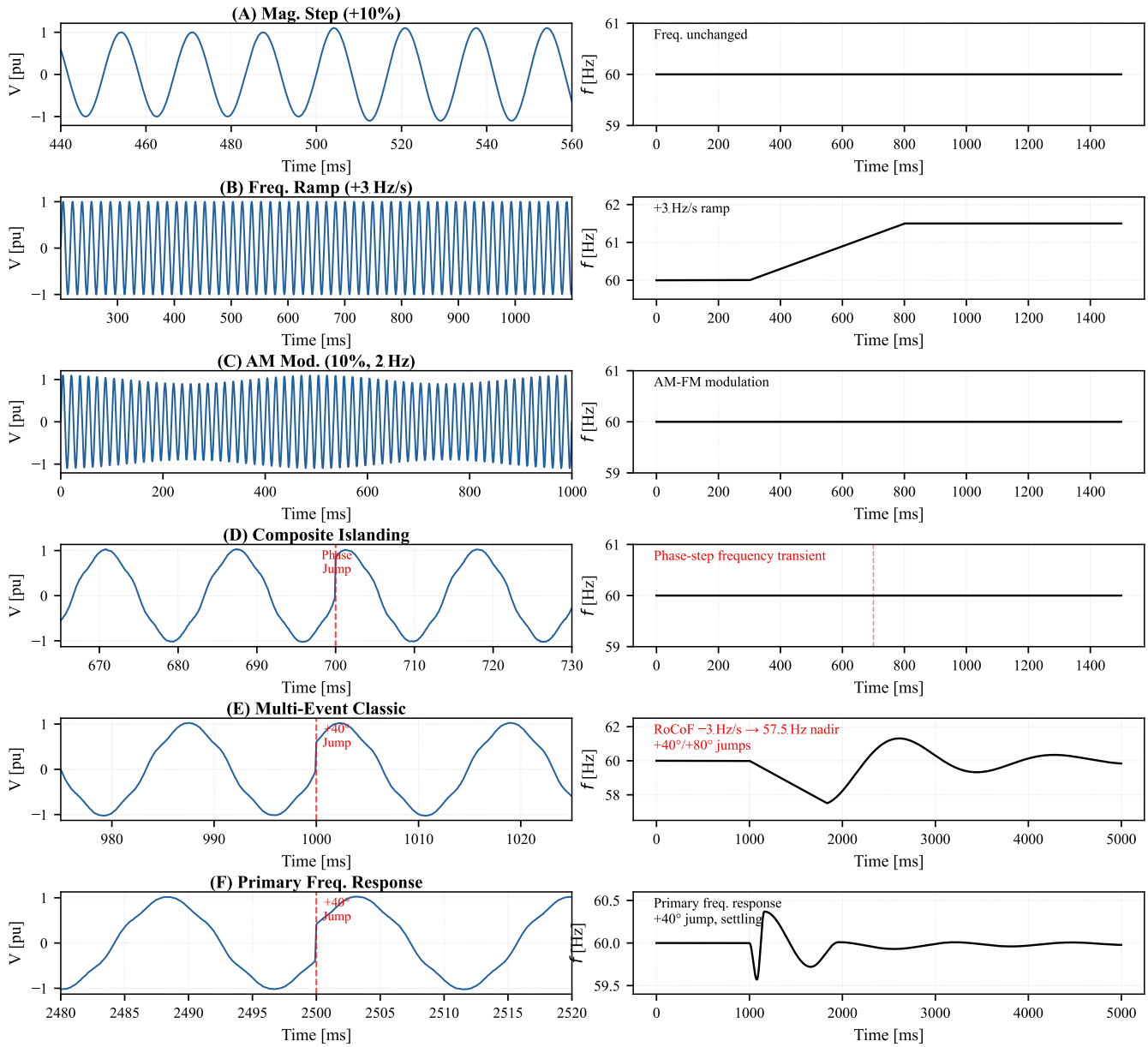

Fig. 1: Complete Stress-Test Scenario Suite. (A–C) Dynamic performance profiles following the test philosophy of IEC/IEEE 60255-118-1 (step, ramp, modulation). (D) Composite Islanding featuring an instantaneous $+60^\circ$ phase jump with harmonics and elevated noise ($\sigma = 0.005$ pu; Section II-B). (E) IBR Multi-Event: scenario-dependent noise ($\sigma_{\text{base}} = 0.003$ pu plus sparse impulsive spikes ≤ 0.05 pu); frequency profile shows the full 5 s composite sequence.

RMSE < 0.300 mHz. **Ramp:** The Ramp test exposes structural limitations. SOGI-FLL fails to track (RMSE $=$ 1120 mHz; type-1 loop). Among evaluated estimators under this scenario, the RA-EKF achieves RMSE $=$ **8.23** mHz—8.15× lower than SRF-PLL and 1.41× lower than EKF.

B. Robustness Under Severe IBR Disturbances

1) Composite Islanding (Scenario D)

Window-based methods exhibit spectral leakage: IpDFT reaches RMSE $=$ 484 mHz (peak 2910 mHz); TFT peak 2740 mHz. EKF/UKF, lacking phase-reset mechanisms, incur $T_{\text{trip}} = 0.122$ s; SRF-PLL avoids leakage (RMSE $=$ 45.0 mHz, $T_{\text{trip}} = 0$). Among model-based estimators, the RA-EKF achieves RMSE $=$ 39.3 mHz—a 5.7× RMSE reduction vs. EKF—with zero threshold breaches ($T_{\text{trip}} = 0$; peak 365 mHz $<$ 0.5 Hz, Fig. 2g). *Note:* Scenario D applies all disturbances simultaneously; isolating the contribution of harmonic leakage vs. noise amplification requires single-effect experiments outside this benchmark's scope.

2) Multi-Event Stress Test (Scenario E)

Table IV quantifies trip-risk under the IBR Multi-Event sequence (5th/7th harmonics; THD $\approx$ 4.5%, IEEE 519 compliant). SOGI-FLL incurs the highest trip-risk (3.10 s); SRF-

TABLE III: Unified Benchmark Results: Standard vs. IBR-Specific Stress Scenarios [Errors in **mHz**, T_{trip} in **s**; mean $\pm$ std, $N = 30$].

	IEC/IEEE 60255-118-1 (Standard Scenarios)						Composite Islanding (Scenario D)	
	Step (+10%)		Ramp (+3 Hz/s)		Modulation (2 Hz)		Phase Jump (+60°)	
Method	**RMSE**	**Peak**	**RMSE**	**Peak**	**RMSE**	**Peak**	**RMSE [mHz]**	T_{trip} **[s]**
Window-based								
IpDFT	1.60 ± 0.04	7.53 ± 0.23	35.3 ± 0.02	83.0 ± 1.38	0.260 ± 0.02	0.680 ± 0.10	484 ± 0	0.086
TFT	2.12 ± 0.03	19.0 ± 0.23	30.7 ± 0.02	55.7 ± 0.67	0.260 ± 0.02	0.820 ± 0.13	479 ± 0	0.087
Data-driven								
Koopman	1.25 ± 0.04	8.79 ± 0.21	28.6 ± 0.02	54.7 ± 0.97	1.23 ± 0.02	2.20 ± 0.12	419 ± 0	0.148
Loop-based								
SRF-PLL	**0.140**	**0.730**	67.1 ± 0.02	173 ± 0.28	**0.130**	**0.250**	45.0 ± 0	**0.000**
SOGI-FLL	8.99 ± 0	13.4 ± 0	$1120^{\dagger}$	1560 ± 0.03	8.99 ± 0	13.4 ± 0	**8.36**	**0.000**
Model-based								
EKF	0.240 ± 0.01	2.52 ± 0.03	11.6 ± 0.11	41.1 ± 3.08	0.150 ± 0	0.570 ± 0.06	225 ± 0	0.120
UKF	0.280 ± 0.04	2.57 ± 0.04	12.7 ± 0.16	45.2 ± 3.70	0.280 ± 0.02	1.70 ± 0.16	225 ± 0	0.120
RA-EKF	2.31 ± 0.02	29.9 ± 0.24	**8.23**	**31.8**	4.78 ± 0.03	12.8 ± 0.41	39.3 ± 2.40	**0.000**

†SOGI-FLL fails to track sustained RoCoF (type-1 frequency loop). All errors in mHz. ± 0: std < 0.5 mHz (negligible MC variance; deterministic scenario response). Bold: best in category.

PLL accumulates $T_{\mathrm{trip}} = 0.353$ s (RMSE $= 307$ mHz). The RA-EKF achieves RMSE $= 159$ mHz with $T_{\mathrm{trip}} = 0.081$ s at $88.5\,\mu$s/sample—a $4.4\times$ trip-risk reduction vs. SRF-PLL and $4.5\times$ vs. EKF in this benchmark. IpDFT: $T_{\mathrm{trip}} = 0.563$ s; Koopman: 0.068 s.

C. Computational Feasibility

Table V and Fig. 2e show the cost–risk Pareto frontier. All timings are Python 3.13 baselines (Intel Core Ultra 5 125H, single-threaded) for *relative comparison only*; C/FPGA implementations yield 10–100$\times$ lower absolute latencies. SRF-PLL ($16.7\,\mu$s) and EKF ($29.5\,\mu$s) are least expensive; SRF-PLL incurs elevated trip-risk ($T_{\mathrm{trip}} = 0.353$ s, Scenario E). RA-EKF ($88.5\,\mu$s) is a viable DSP/FPGA candidate pending C/HIL validation. PI-GRU ($\approx$154 ms/sample) is prohibitive for embedded real-time protection without GPU acceleration.

D. Failure Mode Summary by Estimator Family

Dominant failure modes (Table I): **Window-based:** spectral leakage ($>$ 450 mHz RMSE, Scenario D) and structural latency. **Loop-based:** SRF-PLL cycle-slip under phase jumps; SOGI-FLL type-1 loop fails sustained RoCoF (1120 mHz, Scenario B). **Adaptive recursive:** VFF blow-up and covariance instability under multi-event transients; TKEO diverges under noise (RMSE ≥ 7800 mHz, excluded from Tables II–III). **Data-driven:** PI-GRU dataset shift—the fixed pre-trained model underperforms scenario-optimized classical methods on RMSE and trip-risk in all five scenarios. **Model-based:** EKF/UKF Jacobian divergence ($T_{\mathrm{trip}} = 0.120$ s, Scenario D); RA-EKF event-gating mitigates this.

TABLE IV: Trip-Risk Analysis: Multi-Event Scenario (E). T_{trip}: total cumulative time summed over all non-contiguous intervals where $|e(t)| > 0.5$ Hz (0.5 Hz threshold from IEC/IEEE 60255-118-1 Class 0 RMSE accuracy class, used here as a protection-relevant trip proxy for visualisation only; no formal IEC compliance is claimed). Note: multiple short excursions may be more harmful than a single long one in a relay-protection context; T_{trip} does not capture excursion count or inter-excursion timing. CPU: mean time per sample (μs), Scenario E run (Python baseline; not a real-time deployment projection). Values: mean $\pm$ std from Monte Carlo (N=30). Methods not shown: TKEO (RMSE ≥ 7800 mHz, diverges); RLS (342 mHz, $T_{\mathrm{trip}} = 0.367$ s, CPU $= 12.7\,\mu$s; similar to EKF); RLS-VFF (874 mHz, $T_{\mathrm{trip}} = 2.20$ s); TFT ($T_{\mathrm{trip}} = 0.349$ s); PI-GRU (1830 mHz, $T_{\mathrm{trip}} = 3.76$ s, CPU $\approx 1.54 \times 10^5\,\mu$s; Table V).

Method	**RMSE [mHz]** ($\mu \pm \sigma$)	T_{trip} **[s]** ($\mu \pm \sigma$)	**CPU [μs]**
RA-EKF	159 ± 3	0.081 ± 0.001	88.5
EKF	335 ± 0	0.364 ± 0.001	29.5
Koopman	471 ± 0	0.068 ± 0.000	120
UKF	337 ± 0	0.366 ± 0.001	215
SRF-PLL	307 ± 0	0.353 ± 0.001	16.7
IpDFT	636 ± 0	0.563 ± 0.000	110
SOGI-FLL	826 ± 0	3.100 ± 0.000	41.7

TABLE V: Computational Cost per Sample (Python Baselines). CPU: mean time per sample over 20 runs on an Intel Core Ultra 5 125H (single-threaded Python 3.13; not C or FPGA projections). Deployment class: P1 ($<20\,\mu$s, embedded MCU), P2 (20–40 μs, DSP), M1 ($>40\,\mu$s, DSP/FPGA or higher); these are per-sample execution time thresholds, not estimator response-latency classes.

Method	**Time/Sample (μs)**	**Class**	**Target HW**
SRF-PLL	$16.7\,\mu$s	P1	Low-end MCU
EKF	$29.5\,\mu$s	P2	DSP
SOGI-FLL	$41.7\,\mu$s	M1	MCU / DSP
TFT	$45.6\,\mu$s	M1	DSP
RA-EKF	$88.5\,\mu$s	M1	DSP / FPGA
IpDFT	$110\,\mu$s	M1	DSP / FPGA
Koopman	$120\,\mu$s	M1	CPU
UKF	$215\,\mu$s	M1	High-perf CPU
PI-GRU	$153{,}900\,\mu$s ($\approx$154 ms)	—	GPU only

IV. Conclusion

The simulation results indicate that standard IEC/IEEE 60255-118-1 evaluation tests may not fully characterize frequency estimator behavior in converter-dominated grids, where phase discontinuities, high RoCoF, and harmonic distortion can expose failure modes that isolated single-disturbance tests do not reveal. This benchmark evaluates twelve estimators across five algorithmic families under five stress scenarios and reports three main observations.

1) **Latency–Robustness Trade-off:** Window-based methods achieve strong steady-state accuracy, but their behavior differs across scenario types. IpDFT is better suited to metering-oriented operation, whereas TFT reduces observation delay at the cost of higher noise sensitivity under composite disturbances. Loop-based estimators (SRF-PLL,

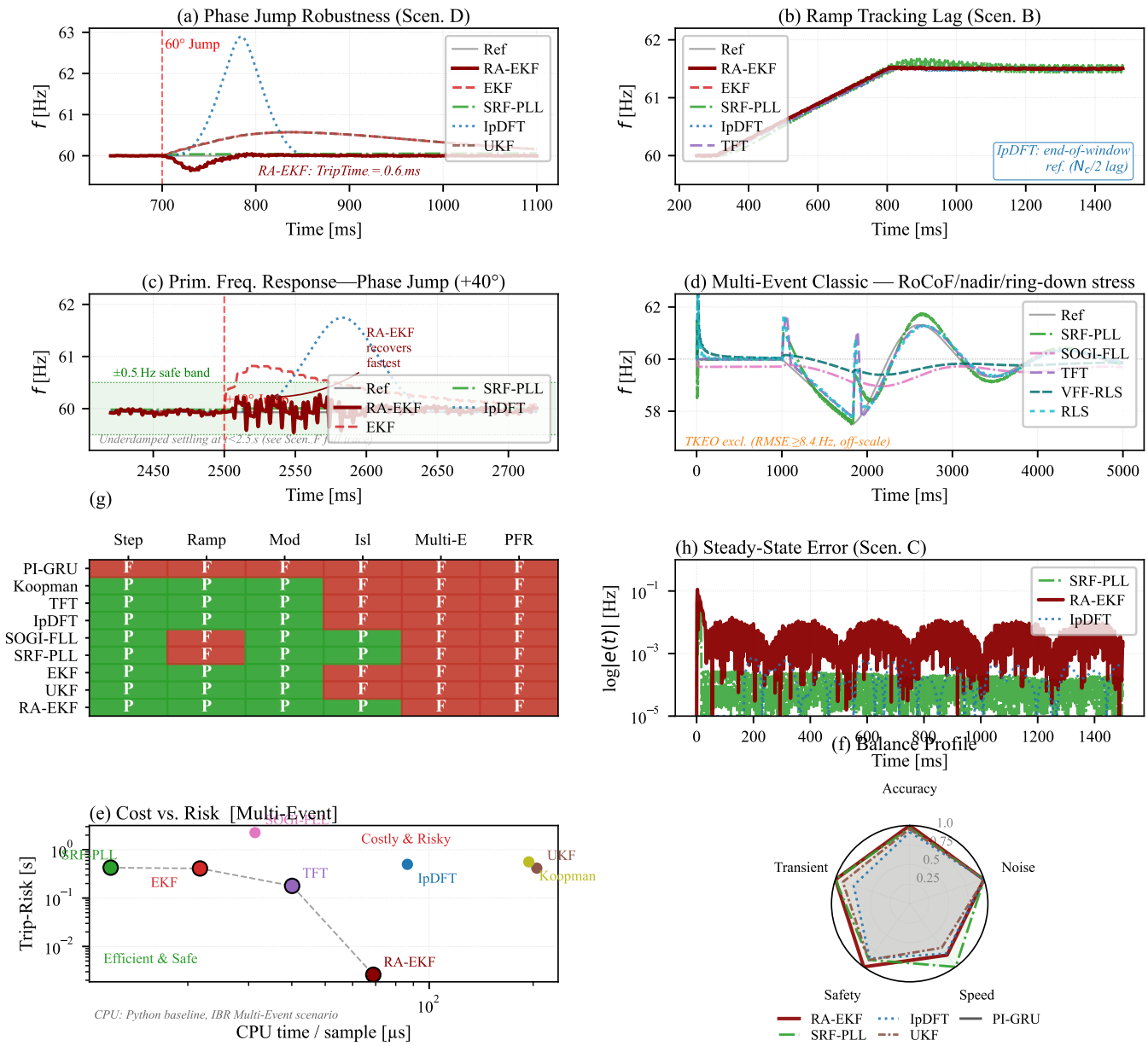

Fig. 2: Comprehensive Performance Dashboard (all errors in mHz; $N = 30$ MC). **(a)** Time-domain frequency error, Scenario D (Composite Islanding): event-gating in RA-EKF limits T_{trip} to ≈0.40 ms vs. 122 ms for standard EKF. **(b)** Time-domain frequency error, Scenario B (Ramp): RA-EKF lowest RMSE among evaluated estimators (8.23 mHz) via explicit $\dot{\omega}$ state; IpDFT at 2-cycle window has 33 ms structural latency (observation delay, not stochastic lag) and achieves RMSE = 35 mHz; SOGI-FLL fails (1120 mHz) due to type-1 loop. **(c)** Time-domain, Scenario E: stable estimators. **(d)** Time-domain, Scenario E: unstable estimators (TKEO excluded, RMSE ≥ 7800 mHz). **(e)** Pareto frontier: CPU time/sample vs. T_{trip}, Scenario E; 2D projection only—RMSE is a third independent dimension not shown here. †PI-GRU excluded (CPU ≈154 ms/sample, $T_{\text{trip}} \approx 3.8$ s). All CPU values are Python baselines; embedded speedups of 10–100× expected with C/FPGA implementations. **(f)** Radar chart: normalised multi-metric profile per estimator (lower area = better overall balance across all metrics). **(g)** Non-normative compliance heatmap, all scenarios. Pass thresholds: RMSE < 0.05 Hz, Peak < 0.5 Hz, $T_{\text{trip}} < 0.1$ s (from IEC/IEEE 60255-118-1 accuracy classes; visualisation only, no certification claim). **F*** = marginal failure: RMSE overshoot < 1.5% or sub-cycle peak breach only, compliant in all other metrics. RA-EKF, Scenario D: peak = 365 mHz, $T_{\text{trip}} = 0$. **(h)** Steady-state RMSE, log scale, Scenario C (Modulation); all values < 1 mHz for top-performing estimators.

SOGI-FLL) are computationally efficient and handle isolated phase jumps well, but exhibit elevated trip-risk under the composite Multi-Event sequence ($T_{\text{trip}} = 0.353$ s for SRF-PLL; Table IV).

2) **Model-Based Filters:** Within the model-based family, the RA-EKF (augmented with an explicit $\dot{\omega}$ state, Innovation-Driven Covariance Scaling, and Event Gating) achieves $T_{\text{trip}} = 0$ under Composite Islanding vs. $T_{\text{trip}} = 0.120$ s for

the standard EKF ($N = 30$ Monte Carlo), at $88.5\,\mu$s/sample (Python baseline; C/HIL validation remains future work). The standard EKF and UKF offer lower cost but higher trip-risk under composite disturbances; all three model-based filters require careful tuning for IBR-specific disturbance types.

3) **Limits of Standard Compliance Tests:** Several estimators passing IEC/IEEE 60255-118-1 step and ramp tests exhibited degraded performance under composite multi-disturbance scenarios, suggesting that standard compliance tests may not fully capture estimator resilience in IBR-dominated conditions.

Data-driven approaches show variable cost: PI-GRU requires ≈ 154 ms/sample (Python/PyTorch CPU baseline), prohibitive for embedded protection without GPU acceleration, and did not outperform scenario-optimized classical methods in any of the five evaluated scenarios. This benchmark is simulation-based; hardware-in-the-loop validation remains a next step. The results suggest that IEC/IEEE 60255-118-1 compliance tests do not fully characterize estimator resilience in IBR-dominated grids; a *Composite Phase-Step with Harmonic Distortion* scenario—analogous to Scenarios D–E—with probabilistic compliance bounds ($\mathrm{PCB} = \mu_{|e|} + 3\sigma_{|e|}$) may better assess relay security. Simulation code and hyperparameter grids are available from the authors for reproducibility.

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